Beyond Planted Strings: Semantic Canaries, Canary Detectability, and Partial Unlearning in Anytime-Valid Unlearning Certificates

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Abstract

VOUCH certifies machine unlearning with paired ghost canaries: a fair coin decides which of two exchangeable records is trained and later unlearned, which makes the sign of their score difference an exact fair coin under exact unlearning, and betting e-processes turn these signs into an anytime-valid certificate on the committed cohort. The certificate was limited to random-string canaries and to a provider trusted not to treat canaries differently from other data. We extend it in three directions. First, semantic twins, two incompatible natural-language facts about one fictitious entity, keep the exact null for every score that compares the two facts under the same prompt, including belief scores over paraphrased questions never seen in training. Second, under a bounded-influence assumption the cohort certificate transfers to the canary generator's population with a McDiarmid term, and from canaries to organic records with a term equal to twice the provider's ability to tell them apart from their contents, which prices content-based special-casing, and for one-sided leakage the bound is exact and equals a Neyman–Pearson power. VOUCH's random-string canaries and naive semantic twins are told apart with advantage one; twins drawn from the organic record generator are not, but on Pythia-410M at our training budget a record seen once leaves no detectable trace, so indistinguishability costs the audit its power. Third, on Pythia-410M and Pythia-1B, when unlearning targets only the literal statement, the advantage on paraphrased questions is retained slightly longer than the literal one (**+0.07** to **+0.12** in like-for-like retention), consistently in direction but not significantly at eight seeds. When the forget set is incomplete, an audit that adds calibrated paraphrase scores revokes up to **38%** of runs that a literal-only audit certifies, driven by a minority of seeds; when it is complete, the two audits agree. On the same runs, a membership-inference test against an organic holdout raises a false alarm on every retrained model, and against matched ghost twins it is valid but far less powerful than a paired test on the

same records, which is where VOUCH's advantage comes from. In the sign model we show that no sequential test needs fewer than $\mathbf{\log(1/\alpha)/KL}$ samples and that an oracle bettor comes within a logarithmic term of that, comparable to a fixed-sample sign test at **78%** power, and we bound the cohort's certification time through the growth of the wealth process.

Keywords: machine unlearning, anytime-valid inference, e-values, canaries, membership inference, large language models

1 Introduction

A user who asks a model provider to delete their data receives, at best, an assurance that an unlearning routine was run. Whether the routine removed what it was meant to remove is a statistical question, and VOUCH [1] gave it a statistical answer. Before fine-tuning, the provider plants *paired ghost canaries*: pairs of exchangeable records of which a fair coin admits one into training and, later, into the forget set, while the other is never trained on. Under exact unlearning the deployed model cannot depend on the coins, so the sign of the score difference between the two twins is a fair coin, exactly and at any sample size. Betting e-processes built on these signs give an anytime-valid (ε, α)-forgetting certificate, a revocation alarm, and a confidence sequence on the residual advantage, with no shadow models and no retrained reference.

The certificate VOUCH issues has a precise scope: it bounds the realised advantage of a declared class of scores on the committed cohort of canaries, under a provider that runs the same pipeline on canaries and on everything else. That scope leaves three questions open, and each has practical weight.

First, do canaries stand in for the data users delete? VOUCH's canaries are templated records with random secret strings. Users delete biographies, messages and facts. A certificate on random strings says nothing about natural-language facts unless something connects the two, and the connection has to survive a provider who might, deliberately or not, treat the two differently.

Second, what does a strategic provider need in order to pass the audit without unlearning? The honest-but-verifiable threat model of VOUCH assumes the provider does not special-case the cohort. The assumption is reasonable for an audit run with the provider's cooperation, but it is an assumption, and the original study measured that its own canary templates are easy to recognise. How much can a provider gain by recognising canaries, and what would it take to deny it that gain?

Third, does an audit of the trained form see knowledge that survives in other forms? Gradient-based unlearning has been reported to suppress the exact string it was given while the fact it encodes stays reachable through a paraphrase [2], and a deletion request may name one record of a fact the model learned from several. A random-string secret has no paraphrase, so VOUCH's canaries cannot probe this failure. Semantic canaries can, provided the exact null survives the move to natural language.

Contributions.

This paper extends VOUCH along these three lines, with new theory and new experiments on Pythia models [3].

- **Semantic twins keep the exact null.** A twin pair that states two incompatible facts about one fictitious entity keeps VOUCH's exact conditional null for every score that compares the two facts under the same prompt, including belief scores over paraphrased questions that never appear in training, and their versions calibrated by the base model (Proposition 1). This realises the natural-language canary that the original paper named as its next step.
- **A ladder from the cohort to organic records.** Under a bounded-influence assumption, the cohort certificate transfers to the canary generator's population with a McDiarmid term (Theorem 3); and from canaries to organic records with a term equal to twice the provider's ability to tell the two apart from their contents (Theorem 4). The second result prices one deviation from the honest-provider assumption, content-based special-casing: whatever rule a provider uses to treat canaries differently, the gap it opens is bounded by that rule's distinguishing advantage (Corollary 5). For one-sided leakage the bound is exact: the largest organic advantage compatible with a certified cohort is the Neyman–Pearson power of testing canaries against organic records at level ε, which is ε when the two coincide and 1 when they are mutually singular (Theorem 6). Any detector therefore gives a measurable lower bound on what a selective provider can leave un-unlearned while the audit passes.
- **Detectability and its price, measured.** VOUCH's random-string records and semantic twins built from their own templates are detected with advantage one by a character n-gram classifier. Twins drawn from the organic record generator itself, with one slot varied, are not detected by perplexity or n-gram detectors beyond the noise of the detectors. But when such records appear once, as organic records do, Pythia-410M retains no detectable trace of them at our training budget, so the audit becomes valid but powerless: the term the ladder pays for detectability is traded for power (Sections 8.2 and 8.5).
- **The literal form and the fact under partial unlearning.** When unlearning targets only the literal statement, the advantage on paraphrased questions is retained slightly longer than the literal one in all three experiments we ran (two model sizes, two designs), by 0.07 to 0.12 in like-for-like retention; the direction is consistent but no test reaches significance at eight seeds, so we do not claim to have demonstrated the squeezing effect at this scale. What does change verdicts is an incomplete forget set: when the fact was also trained in a form that the forget set does not contain, a union audit over literal and calibrated paraphrase scores revokes up to 38% of runs that a literal-only audit certifies, an effect driven by a minority of seeds and close to true by construction, since the paraphrase scores probe a form the forget set omitted; when the forget set is complete the union agrees with the literal audit and keeps its error rate (Section 8.4).
- **Comparison with membership-inference audits.** On the same runs, a membership-inference test of canaries against an organic holdout raises a false alarm on every retrained model, because the test detects the difference between canaries

and organic records rather than membership; against matched ghost twins it is valid but misses partial unlearning that a paired test on the same records detects; a paired fixed-sample sign test is about as powerful as VOUCH, so the advantage comes from pairing, and sequential testing buys continuous monitoring at some cost in power (Section 8.6).

- **What enlarging the class and monitoring continuously cost.** Adding scores to the declared class costs nothing in error on the certificate side and the logarithm of a weight on the revocation side (Proposition 2). In the sign model, no level-α sequential test beats $\log(1/\alpha)/\mathrm{KL}$ expected samples, and an oracle constant-stake bettor comes within $\log(1+\Delta)/\mathrm{KL}$ of it, which equals the sample size of a fixed-sample sign test with about 78% power at $\alpha = 0.05$ (Proposition 8); the ten-stake mixture of the engine costs 1.1 to 1.4 times more. This explains why a paired fixed-sample test is as powerful as VOUCH on the same pairs.
- **When the cohort certificate arrives.** Under a uniformly random reveal order the betting edge of VOUCH's without-replacement process is deterministic (Lemma 9). A Freedman bound on the wealth process then gives a tail bound on the certification time that holds before refutation and shows where the finite-population speed-up comes from (Theorem 10); it is checked numerically and is conservative by a factor of two to four.

All code, results and the scripts that turn results into the tables of this paper are released with the VOUCH repository.

What is not claimed.

The organic-record result is conditional on the distinguishing advantage, which data can bound only from below, and on a bounded-influence assumption that cannot be checked; we report the strongest detectors we could build, not a proof that none better exists. Squeezing is not demonstrated. All facts in the experiments are fictitious, the data come from one domain, and the models have at most one billion parameters and are fine-tuned with LoRA. These limits are discussed in Section 9.

Organisation.

Section 2 recalls VOUCH and fixes notation. Section 3 places the work in the literature. Sections 4–7 develop the theory: semantic twins and the cost of enlarging the class, the ladder to organic records and its sharp form, the price of anytime validity, and the certification time of the cohort. Section 8 reports the experiments and Section 9 discusses limitations.

2 Background: the VOUCH certificate

This section recalls the parts of VOUCH [1] that the extension builds on and fixes notation; proofs are in the original paper.

Pipeline and threat model.

A provider fine-tunes a model on a corpus, receives deletion requests, applies an unlearning routine to a forget set, and deploys the result M. An auditor (the verifier)

wants evidence that the routine removed the influence of the forget set. The provider is *honest but verifiable*: it runs the committed pipeline on the committed data, and in particular processes the canaries exactly as it processes the rest of the forget set, but it is not trusted to report whether unlearning succeeded.

Paired ghost canaries.

Before fine-tuning, the verifier generates a cohort of m pairs with contents $C_i = (c_i^0, c_i^1)$ and draws independent fair coins $b_i \sim \mathrm{Bern}(\frac{1}{2})$. The in-twin $c_i^{b_i}$ is inserted into the training corpus and later into the forget set; the ghost twin $c_i^{1-b_i}$ is never trained on. The manifest of contents and coins is committed (VOUCH uses a Merkle commitment), and each coin is opened only when its pair is audited. All remaining randomness of training and unlearning is collected in ξ.

Scores, signs and the certified quantity.

A score $s(M, c)$ measures how much the model favours a record, for example the mean token log-probability of a secret after its prompt. A declared class $\mathcal{F}$ of scores is fixed before any query. For each pair and score the verifier computes the difference $D_i^{(s)} = s(M, c_i^{\mathrm{in}}) - s(M, c_i^{\mathrm{ghost}})$ and the sign $Z_i^{(s)} = \mathbf{1}[D_i^{(s)} > 0]$, ties split by a committed coin. The certified quantity is the realised cohort advantage

$$\Delta^{(s)} \;=\; \frac{2}{m} \sum_{i=1}^{m} Z_i^{(s)} \;-\; 1,$$

which is zero in expectation under exact unlearning and positive when the in-twins are still favoured. Negative values (the in-twin scores conspicuously below its partner) are leakage too, and VOUCH certifies both directions.

The exact null.

If M is conditionally independent of the coins given the contents, which holds for exact unlearning and for retraining from scratch, then for every score and every step t of a reveal order chosen independently of the signs, the sign of the pair opened at step t is a fair coin given everything opened before (VOUCH Theorem 1). No independence across pairs is needed: the signs share one model, but the coin governing each pair is fair given the past.

The certificate.

The verifier reveals the cohort in a uniformly random order π, drawn independently of the signs (VOUCH recommends a public randomness beacon), and for each score and direction runs a betting wealth process $W_t = \prod_{u \le t} \big(1 + \lambda_u (p_0(u) - Z_{\pi(u)})\big)$ against the *without-replacement boundary*

$$p_0(t) \;=\; \frac{m p_0 - S_{t-1}}{m - t + 1}, \qquad p_0 = \tfrac{1}{2} + \tfrac{\varepsilon}{2}, \tag{1}$$

where S_{t-1} is the number of ones revealed before step t. If the cohort advantage were at least ε, the conditional mean of the next sign would be at least $p_0(t)$, so W_t is a nonnegative supermartingale and Ville's inequality bounds the probability that it ever reaches $1/\alpha$. A two-sided (ε, α)-certificate is issued when every per-score, per-direction process has crossed $1/\alpha$, and then Pr[a false certificate is ever issued] $\leq \alpha$ under arbitrary dependence between pairs (VOUCH Theorem 2). If $p_0(t) > 1$ the null is infeasible and is refuted outright. A separate revocation process tests the exact null directly and raises an alarm (status REVOKED) when it accumulates evidence that the model still depends on the coins. A run that neither certifies nor revokes ends UNDETERMINED.

Transfer to the coin-averaged advantage.

The cohort advantage depends on the realised coins. If flipping one coin changes at most κ signs, McDiarmid's inequality [4] bounds the distance to the advantage averaged over the coins with the contents held fixed by $\kappa\sqrt{2\log(2/\alpha')/m}$ (VOUCH Proposition 3). Section 5 extends this transfer to the contents and then to organic records.

Notation for this paper.

We write Δ for the cohort advantage of a fixed score when the score is clear, $\mathcal{F}_t$ for the verifier's filtration after t reveals, and $\mathbb{E}$ for expectation. "Literal" refers to scores computed on the trained surface form of a canary, "paraphrase" to scores computed on questions never seen in training.

3 Related work

Unlearning methods and their evaluation.

Approximate unlearning for language models is dominated by gradient ascent on the forget set, negative preference optimisation [5] and its reference-free variant SimNPO [6], and representation misdirection [7]. They are evaluated on benchmarks such as TOFU [8] and MUSE [9] whose forget-quality and leakage metrics are fixed-sample statistics compared against a retrained model. A growing literature shows that these metrics overstate forgetting: forget-quality metrics track surface behaviour rather than retained knowledge [10], output matching hides degenerate models [11], benign relearning restores erased content [12], relearning attacks can be formalised and partly resisted [13], quantisation brings forgotten knowledge back [14], lightweight attacks recover unlearned capabilities [15], and current evaluations can be made to overstate or understate success [16]; see Liu et al. [17] and Xue et al. [18] for surveys. Closest to our partial-unlearning experiment, Li et al. [2] show that gradient-based unlearning shifts probability mass onto paraphrases of the targeted answer, which they call squeezing, and propose an objective that accounts for the model's own beliefs. We study the measurement side of the same phenomenon: whether a certificate on the trained form can be misled, and what an audit needs to see it.

Membership inference as an audit.

Membership-inference attacks are the usual instrument of unlearning audits. Score-based attacks threshold the loss, its zlib-normalised version [19], or the mean of the least likely tokens, Min-K% Prob [20]; reference-based attacks calibrate the loss by a reference model, and LiRA [21] calibrates per example with shadow models. Hayes et al. [22] show that unlearning evaluations based on population-level attacks overstate privacy and that per-example calibration with shadow models is needed. Zhang et al. [23] establish that membership-inference scores cannot prove that a model was trained on given data unless the inclusion of that data was randomised, the principle behind VOUCH's fair coins. For pre-trained language models, Duan et al. [24] find that membership-inference attacks perform near chance when members and non-members are drawn from the same distribution, and Das et al. [25] show that classifiers that never query the model match or beat published attacks on benchmarks whose members and non-members differ in distribution. Our matched- and unmatched-holdout baselines (Section 8.6) show what happens to a membership-inference audit when that principle is not respected, and Theorem 4 states the same confound in the language of a certificate.

Verification of unlearning.

Thudi et al. [26] argue that approximate unlearning cannot be audited from weights alone and that auditable definitions are a prerequisite for any guarantee; Zhang et al. [27] show that a dishonest provider can defeat existing families of verification strategies while passing them; cryptographic proof-of-unlearning [28] attests that the committed computation ran, which is complementary to whether forgetting holds statistically. VOUCH [1] made statistical verification anytime-valid with paired canaries and betting e-processes [29–32], building on the secret-sharer methodology [33]. Privacy auditing with canaries [34] shares the planted design but estimates a lower bound on the privacy parameter of a training algorithm, not the outcome of one unlearning run. Copyright traps [35] plant sequences in documents so that their inclusion in a model's training data can be tested later, a planted design closer in spirit to ours, without a paired control. Yu et al. [36] show that retraining-based definitions of unlearning are not well posed for multi-stage pipelines, one reason VOUCH avoids a retrained reference. The special-casing provider that VOUCH's threat model excludes is the subject of Section 5: we do not remove the assumption, but we price it.

What is new here.

Relative to VOUCH, this paper (i) moves the canaries to natural language and proves that the exact null survives; (ii) states and proves the transfer from canaries to organic records as a function of canary detectability, measures detectability for several designs including one built to make it negligible, and measures what that design costs in power; (iii) tests the certificate against unlearning that targets only the literal form, in two designs and at two model sizes, and evaluates a union audit with paraphrase scores; (iv) compares with membership-inference audits on the same runs, with retraining as an exact-null control; and (v) bounds the certification time through wealth growth rather than through refutation.

4 Semantic twins

VOUCH's secrets are random alphanumeric strings. A high-entropy string has no semantically equivalent rephrasing, so a score that reads the secret under paraphrased carrier sentences finds the same advantage as the literal score, and the squeezing effect of Li et al. [2] cannot be observed; the original paper reports exactly this and names natural-language secrets, exchangeable as propositions rather than as strings, as the next step [1]. This section defines such canaries and shows that the exact null survives.

4.1 The generator

A semantic twin pair names a fresh fictitious entity e and an attribute a (for example birthplace or profession) and states two incompatible values of it: c^0 says that e has value v, c^1 that e has value $v' \neq v$. The pair is drawn by a generator controlled by the verifier.

Assumption 1 (Generator) The pair contents $C_1, \ldots, C_m$ are independent draws from a generator distribution P on ordered pairs that is *label-symmetric*: $(c^0, c^1) \sim P$ implies $(c^1, c^0) \sim P$. The coins b_i are independent Bern($\frac{1}{2}$) draws, independent of the contents and of ξ, and ξ is independent of the contents.

Both properties are under the verifier's control, because they describe its own sampler rather than the data or the model. Our generator draws the two values as an unordered pair from the attribute's value set and swaps them with probability one half, which makes it label-symmetric by construction; the coins come from a separate random stream. The in-twin is trained in one or more *training forms*, for example the literal statement "e was born in v" and one question–answer form.

4.2 Scores

Each score compares the two values under the same prompt, so that both twins of a pair are scored by the same function of the model:

- *literal*: the mean token log-probability of the value after the literal prefix ("e was born in"), $s_{\text{lit}}(M, c^k) = \overline{\log P_M}(v_k \mid \text{prefix})$;
- *belief*: the same quantity averaged over a committed set Q of paraphrased questions ("Where was e born?", "What is the hometown of e?"), none of which appears in training;
- *belief_max*: the maximum over Q instead of the mean, the semantic analogue of VOUCH's s_{para};
- the *calibrated* version of each, which subtracts the same score computed under the frozen base model M_0: $s^{\text{cal}}(M, c) = s(M, c) - s(M_0, c)$.

Calibration removes the base model's prior preference between the two values (a base model may find "born in Paris" more likely than "born in Porto" for any name). It keeps the score in the declared class because M_0 is fixed before the coins are drawn.

In the experiments we refer to the literal score as *literal* and to the calibrated belief scores as *paraphrase*.

4.3 The exact null

Proposition 1 (Exact null for semantic twins) *Under Assumption 1, if the deployed model is conditionally independent of the coins given the contents, then for every score s of the form above and every step t of a reveal order chosen independently of the signs, $Z^{(s)}_{\pi(t)} \mid \mathcal{F}_{t-1} \sim \mathrm{Bern}(\frac{1}{2})$ exactly.*

Proof We follow the proof of VOUCH Theorem 1. Enlarge $\mathcal{F}_{t-1}$ to $\mathcal{G}_{t-1}$, which also reveals the position $\pi(t)$ and the *unordered* contents $\{c^0_j, c^1_j\}$ of the pair $j = \pi(t)$ opened at step t, but not its coin. Given $\mathcal{G}_{t-1}$ and the deployed model, the two numbers $s(M, c^0_j)$ and $s(M, c^1_j)$ are determined, because each score is a function of the model and the record alone: both twins share the literal prefix or the question set Q and differ only in the value evaluated, and the calibrated score subtracts $s(M_0, \cdot)$, where the base model M_0 is fixed before any coin is drawn. The coin b_j is independent of the other coins and of the contents by Assumption 1, independent of M given the contents by hypothesis, and no earlier reveal involves it, so $b_j \mid \mathcal{G}_{t-1} \sim \mathrm{Bern}(\frac{1}{2})$. Since b_j decides which of the two determined numbers is the in-twin's, $D_j \overset{d}{=} -D_j$ given $\mathcal{G}_{t-1}$, and the committed tie-break coin splits an atom at zero equally, so $Z_j \mid \mathcal{G}_{t-1} \sim \mathrm{Bern}(\frac{1}{2})$. Averaging over $\mathcal{G}_{t-1} \supseteq \mathcal{F}_{t-1}$ gives the claim. The semantics of the records never enters. □

Two remarks delimit what the proposition gives. First, *semantics costs power, not validity.* A base-model prior between the two values enters both arms symmetrically and leaves the null exact, but it adds variance to the score differences; calibration removes most of it. Second, the null does not need label symmetry: the coin, not the order of the pair, decides which twin is trained. Label symmetry matters for what the certificate says about records outside the cohort. It makes the two components c^0 and c^1 of a pair identically distributed, so every canary record is a draw from a single record distribution, whichever position it occupies and whichever twin the coin trains, and it is this distribution that Section 5 compares with the records users actually delete.

4.4 Enlarging the declared class

Paraphrase scores enlarge the declared class $\mathcal{F}$. VOUCH's certificate and its revocation alarm test different nulls, and they pay differently for the enlargement.

Proposition 2 (Cost of enlarging the class) *Let $\mathcal{F}' \supseteq \mathcal{F}$ be finite and run VOUCH on $\mathcal{F}'$ with per-score, per-direction wealth processes E_s ($s \in \mathcal{F}'$).*

(i) Certificate. *The false-certificate probability stays at most α without any correction for $|\mathcal{F}'|$ (VOUCH Theorem 2). Enlarging the class can only delay issuance: the certificate is issued at the largest of the arm crossing times.*

(ii) Revocation. *The null of the alarm is simple, every score is an exact fair coin, and the E_s are nonnegative supermartingales with $E_s(0) = 1$ under it. For weights*

$w_s \geq 0$ *with* $\sum_s w_s = 1$*, the average* $E = \sum_s w_s E_s$ *is again one, so revoking when* $E_t \geq 1/\alpha$ *at any time has error at most* α *(Ville). Every Bonferroni rule "revoke when some* $E_s \geq 1/(w_s\alpha)$*" is valid and rejects only if* $E_t \geq 1/\alpha$*; the uniform union at* $\alpha/2$ *per arm of two arms is the case* $w_s = \frac{1}{2}$*.*

(iii) Price. *In the sign model of Section 6, if the leaking score carries weight* w *and the oracle bettor is used on it, the expected revocation time is at most* $\big(\log \frac{1}{w\alpha} + \log(1+\Delta)\big)/\mathrm{KL}\big(\frac{1+\Delta}{2} \,\|\, \frac{1}{2}\big)$*, that is, larger by* $\log(1/w)/\mathrm{KL}$ *than for the leaking score alone.*

Proof (i) The proof of VOUCH Theorem 2 conditions on the sign vectors of all scores and, given a violated tolerance of some score s^*, bounds the probability that E_{s^*} ever reaches $1/\alpha$ by α; the false certificate requires that event, so no union over $\mathcal{F}'$ is needed. Issuance needs every arm to cross, hence the maximum of the crossing times. (ii) A weighted average of supermartingales is a supermartingale [the merging of e-values by averaging; 37]. If $E_s \geq 1/(w_s\alpha)$ then $E \geq w_s E_s \geq 1/\alpha$. (iii) Apply Proposition 8(ii) to wE_{s^*} crossing $1/\alpha$, that is E_{s^*} crossing $1/(w\alpha)$. □

Two consequences matter for the experiments. First, a class enlarged with paraphrase scores costs error rate nothing on the certificate side and a logarithm of the weight on the revocation side. Second, our union audit of Section 8.4 splits α in half on *both* sides; by (i) the certificate arms could run at α, so the issuance rates we report for the union are conservative.

5 From the cohort to organic records

The VOUCH certificate concerns the realised advantage Δ on one committed cohort. This section climbs a ladder of three targets: the cohort, the generator population, and organic records, the data users actually delete. Each step costs one term in the tolerance, and each term has its own cause.

5.1 From the cohort to the generator population

VOUCH Proposition 3 averages over the coins with the contents fixed. The verifier also controls the contents, through Assumption 1, so the same argument reaches them.

Assumption 2 (Bounded influence) For fixed ξ, changing a single coin b_i changes at most κ of the m signs, and replacing the contents C_i of a single pair changes at most ℓ of them.

Under exact unlearning $\kappa = \ell = 1$: the model carries no trace of the forget set, so a pair's coin or contents can affect only that pair's own sign. Under partial unlearning κ and ℓ measure how far one canary's training reaches into the others' scores. The assumption is a worst-case bound that the verifier cannot check: with stochastic training, changing one training record perturbs the weights and can flip the sign of every pair whose score difference is close to zero. It is plausible for canaries about distinct fictitious entities, whose score differences are dominated by their own training, but we

state Theorem 3 as a conditional result and use it only to explain how each term of the ladder arises.

Theorem 3 (Transfer to the generator population) *Let $\Delta_\xi = \mathbb{E}_{b,C}[\Delta \mid \xi]$ be the expected advantage of this pipeline run over fresh cohorts from P. Under Assumptions 1 and 2, for any $\alpha' \in (0,1)$, with probability at least $1-\alpha'$ over (b,C),*

$$|\Delta - \Delta_\xi| \;\le\; u_m(\alpha') \;:=\; \sqrt{\frac{2(\kappa^2+\ell^2)\log(2/\alpha')}{m}}.$$

Consequently a two-sided (ε,α) cohort certificate implies $|\Delta_\xi| < \varepsilon + u_m(\alpha')$ with probability at least $1-\alpha-\alpha'$.

Proof Given ξ, Δ is a function of the $2m$ independent coordinates $(b_1,\dots,b_m,C_1,\dots,C_m)$. By Assumption 2, changing one coin changes Δ by at most $2\kappa/m$ and changing one pair's contents by at most $2\ell/m$. McDiarmid's inequality [4] gives $\Pr[|\Delta - \Delta_\xi| \ge u] \le 2\exp(-2u^2/\sum_k c_k^2)$ with $\sum_k c_k^2 = 4(\kappa^2+\ell^2)/m$, that is $2\exp\big(-u^2 m/(2(\kappa^2+\ell^2))\big)$; equating it to α' gives u_m. The consequence follows from a union bound with the certificate's error probability. □

Δ_ξ averages over cohorts but conditions on the training randomness ξ: it is the advantage of *this* deployed pipeline run on the generator population, the relevant target for an auditor of a deployed model. Averaging over ξ would need several independent training runs and is not attempted. At $m = 512$, $\alpha' = 0.05$ and $\kappa = \ell = 1$ the inflation is $u_m = 0.17$, against 0.12 for the coin-only transfer.

5.2 From canaries to organic records

Let P_o be a distribution of *organic* record pairs: two records of the kind a user deletes, with a hypothetical fair coin choosing which one was trained on. For pair contents c, let

$$h_\xi(c) \;=\; \mathbb{E}\big[\,2Z_i - 1 \;\big|\; C_i = c,\ \xi\,\big]$$

be the expected sign contribution of a pair with contents c placed in the cohort, the expectation being over the coins and the other pairs' contents. Then $\Delta_\xi = \mathbb{E}_{c\sim P}[h_\xi(c)]$, and we define the *organic advantage* $\Delta_\xi^o = \mathbb{E}_{c\sim P_o}[h_\xi(c)]$: the advantage the same pipeline run would show on organic pairs planted in the same way. This is the standard notion of membership leakage under randomised inclusion, evaluated on the data users delete; because organic records were not in fact included by a coin, it is a counterfactual quantity of the pipeline, not an observed rate.

Definition 1 (Distinguishing advantage) For a class $\mathcal{H}$ of functions from pair contents to $[0,1]$, $\gamma_{\mathcal{H}}(P,P_o) = \sup_{\phi\in\mathcal{H}}\big|\mathbb{E}_P\phi - \mathbb{E}_{P_o}\phi\big|$. When $\mathcal{H}$ contains all measurable functions, $\gamma_{\mathcal{H}}$ is the total variation distance between P and P_o.

Theorem 4 (Special-casing requires detection) *Let $\phi_\xi = (1+h_\xi)/2$, which takes values in $[0,1]$. Then*

$$|\Delta_\xi^o - \Delta_\xi| \;=\; 2\,\big|\mathbb{E}_P\phi_\xi - \mathbb{E}_{P_o}\phi_\xi\big| \;\le\; 2\,\gamma_{\mathcal{H}}(P,P_o)$$

for every class $\mathcal{H}$ that contains ϕ_ξ. Equivalently, if the pipeline treats canaries and organic records differently enough to open a gap g between their advantages, then ϕ_ξ is a test that tells canary pairs from organic pairs with advantage at least $g/2$.

Proof By linearity, $\Delta_\xi - \Delta_\xi^o = \mathbb{E}_P h_\xi - \mathbb{E}_{P_o} h_\xi = 2(\mathbb{E}_P \phi_\xi - \mathbb{E}_{P_o} \phi_\xi)$, and the right-hand side is at most $2\gamma_\mathcal{H}$ in absolute value whenever $\phi_\xi \in \mathcal{H}$. □

The proof is one line; the content is in the reading. A gap between the canary and the organic advantage of one pipeline run can exist only if the run treats the two kinds of record differently, and ϕ_ξ is the function that describes the difference. The theorem therefore prices one specific deviation from VOUCH's honest-but-verifiable threat model, *content-based special-casing*: a provider that unlearns records more thoroughly when they look like canaries. Whatever rule the provider uses to decide what looks like a canary, the gap it opens is at most twice that rule's advantage at telling canaries from organic records, so a provider that cannot recognise canaries from their contents cannot open a gap at all. Two qualifications apply. The theorem is an upper bound over all pipelines, and Section 5.3 makes it exact for one-sided leakage. And it does not cover special-casing by provenance: if canaries reach the provider through a channel that organic data does not use, the provider can recognise them without looking at their contents, and the verifier must inject canaries through the ordinary data path for the theorem to say anything.

The original VOUCH paper argues that locating the cohort does not threaten the null, because a provider that knows which records are canaries learns nothing about which twin the coin admitted. That argument concerns the validity of the certificate on the cohort and is unaffected here. Theorem 4 concerns a different question, whether the certificate on the cohort says anything about organic records, and there the ability to locate the cohort is exactly what matters. Three consequences follow.

1. If the manifest is committed but its contents are hidden until after deployment, the provider can recognise canaries only from their contents, which is what $\gamma_\mathcal{H}$ measures.
2. An auditor cannot certify $\gamma_\mathcal{H}$ from data. Every detector we train gives a *lower* bound; none gives an upper bound. We therefore report the strongest detectors we could build and state the organic claim conditionally on γ.
3. Making P close to P_o shrinks γ against every detector at once. In the limit where each canary record is drawn from the organic record distribution itself, $\gamma = 0$ at the level of single records.

Corollary 5 (The ladder) *Under the assumptions of VOUCH Theorem 2 and of Theorems 3 and 4, if a two-sided (ε, α) cohort certificate is issued, then with probability at least $1-\alpha-\alpha'$,*

$$|\Delta_\xi^o| \;<\; \varepsilon \;+\; u_m(\alpha') \;+\; 2\gamma_\mathcal{H}(P, P_o).$$

The three terms have separate causes: the tolerance the auditor chose, the size of the cohort, and how far the canaries are from the data they stand in for. Only the third depends on the canary design, and Section 8.2 measures it.

5.3 A sharp form for one-sided leakage

The factor two in Theorem 4 comes from allowing over-forgetting: h_ξ ranges over $[-1, 1]$. When leakage is *one-sided*, meaning no record is forgotten beyond exactness, h_ξ takes values in $[0, 1]$. This covers a provider that skips unlearning on some records, and any pipeline whose unlearning removes only part of the effect of a record, and the bound becomes an equality of optimal values. Let p and p_o be the densities of P and P_o with respect to $\mu = P + P_o$, $\mathrm{TV} = \frac{1}{2}\int |p - p_o|\, d\mu$, and define the *Neyman–Pearson power*

$$\beta(\varepsilon) \;=\; \sup\Big\{ \mathbb{E}_{P_o}\varphi \;:\; \varphi : \mathcal{C} \to [0,1] \text{ measurable},\ \mathbb{E}_P\varphi \le \varepsilon \Big\}.$$

Theorem 6 (Sharp transfer for one-sided leakage) *Let $\varepsilon \in (0, 1]$ and let $B^* = \{p_o > p\}$, $a = P(B^*)$.*

(i) Every pipeline run with $h_\xi \in [0, 1]$ and $\Delta_\xi \le \varepsilon$ has $\Delta^o_\xi \le \beta(\varepsilon) \le \min\{1,\ \varepsilon + \mathrm{TV}(P, P_o)\}$. If, more generally, $h_\xi \in \mathcal{H} \subseteq [0, 1]^{\mathcal{C}}$, then $\Delta^o_\xi \le \Delta_\xi + \gamma_{\mathcal{H}}(P, P_o)$.

(ii) The value $\beta(\varepsilon)$ is attained by a leakage profile $h = \varphi^*$ with φ^* a likelihood-ratio test of P_o against P; hence no argument that uses only the cohort advantage and the generator can exclude $\Delta^o_\xi = \beta(\varepsilon)$. When P has no atoms, φ^* can be taken to be an indicator, which a selective provider realises by skipping unlearning on a set of records.

(iii) $\beta(\varepsilon) \ge \varepsilon\,(1 + \mathrm{TV}/a)$ for $\varepsilon \le a$ and $\beta(\varepsilon) \ge a + \mathrm{TV}$ for $\varepsilon \ge a$. In particular $\beta(\varepsilon) = \varepsilon$ if $P = P_o$, and $\beta(\varepsilon) = 1$ for every $\varepsilon \ge 0$ if P and P_o are mutually singular.

Proof (i) The function $\varphi = h_\xi$ is feasible for the supremum, since $\mathbb{E}_P\varphi = \Delta_\xi \le \varepsilon$ and $\mathbb{E}_{P_o}\varphi = \Delta^o_\xi$. For $\varphi \in [0, 1]$, $\mathbb{E}_{P_o}\varphi \le \mathbb{E}_P\varphi + \mathrm{TV}$, which gives $\beta(\varepsilon) \le \varepsilon + \mathrm{TV}$, and $\mathbb{E}_{P_o}\varphi \le 1$. The statement for a class $\mathcal{H}$ is the definition of $\gamma_{\mathcal{H}}$ applied to $\varphi = h_\xi$. (ii) The Neyman–Pearson lemma [38] gives a (possibly randomised) likelihood-ratio test attaining the supremum; if P is atomless, a non-randomised test of the same size and power exists. (iii) $\varphi = \mathbf{1}_{B^*}$ has $\mathbb{E}_P\varphi = a$ and $\mathbb{E}_{P_o}\varphi = a + \mathrm{TV}$, which is feasible for $\varepsilon \ge a$; for $\varepsilon \le a$ use $(\varepsilon/a)\mathbf{1}_{B^*}$, with $\mathbb{E}_{P_o} = (\varepsilon/a)(a + \mathrm{TV})$. If $P = P_o$ then $\mathrm{TV} = 0$ and (i) gives $\beta(\varepsilon) \le \varepsilon$, while $\varphi \equiv \varepsilon$ attains it. If $P \perp P_o$ then $a = 0$ and $\mathbf{1}_{B^*}$, with $P_o(B^*) = 1$, is feasible for every $\varepsilon \ge 0$. □

Corollary 7 (The ladder, one-sided leakage) *Under the assumptions of Corollary 5, if the leakage is one-sided and a two-sided (ε, α) cohort certificate is issued, then with probability at least $1 - \alpha - \alpha'$, $\Delta^o_\xi < \varepsilon + u_m(\alpha') + \gamma_{\mathcal{H}}(P, P_o)$, and $\Delta^o_\xi \le \beta(\varepsilon + u_m(\alpha'))$ for $\mathcal{H}$ the class of all measurable functions.*

Theorem 6 puts numbers on the two ends of the design space that the experiments explore. For random-string canaries and for semantic twins built from their own templates, P and P_o are separated by a trained detector with advantage one, so $\beta(\varepsilon)$ is close to 1: a selective provider can pass the audit at any tolerance while leaving essentially every organic record un-unlearned, and the certificate says nothing about organic data. For organic-matched twins, $\beta(\varepsilon) = \varepsilon$ and the certificate transfers exactly,

$\Delta^o_\xi \le \varepsilon$ up to the generator term. Between the two, any detector gives a lower bound on $\beta(\varepsilon)$: a provider that unlearns the records its detector calls canary-like and leaves the rest passes the audit when the detector misses at most a fraction ε of canaries, and then leaves a fraction of organic records equal to the detector's true-negative rate at that miss rate un-unlearned. Section 8.2 measures this quantity for our detectors.

Remark 1 (Repetition is part of the content) The provider sees the training corpus, so a canary that is repeated verbatim is recognised by the duplicate detector $\phi(c) = \mathbf{1}[c \text{ occurs at least twice}]$, whose advantage is $P(\text{canary repeated}) - \rho_o$, where ρ_o is the share of organic records that are duplicated. Treating the repetition count as part of a pair's contents, this is a lower bound on γ. Designs that repeat canaries to strengthen memorisation pay for it before any content is examined: our semantic design repeats three quarters of its twins, so $\gamma \ge 0.75 - \rho_o$ with $\rho_o = 0.003$ in our organic corpus, while single-exposure matched twins pay nothing. This is the detectability side of the power trade-off of Section 8.5.

5.4 Organic-matched twins

The third consequence above suggests a design. Let the organic records be produced by a generator G_o, for example a set of sentence templates with slot distributions, and draw each canary pair as follows: sample one record $r \sim G_o$, choose one of its slots (fixed per template), and form the pair by filling that slot with an unordered pair of values drawn from the slot's distribution and swapped by a fair coin. Each twin is then a record of G_o with one slot redrawn, so under a slot distribution that is exchangeable over its values each twin is distributed exactly as an organic record. We call these *organic-matched twins*. Two details matter. Names are drawn as in organic data (no distinguishing middle initial), and each canary appears once, since a record repeated verbatim is itself a signal that organic data rarely gives. The price is power: a record that looks organic is no easier to memorise than an organic record.

Matching is easy here because we wrote the organic generator. For a deployed model the organic distribution is the users' own data, and matching it is the hard part of the problem; Section 9 returns to this.

6 The price of anytime validity

A fixed-sample test is run once, at a sample size fixed in advance, and is invalid if the verifier looks earlier. VOUCH's tests can be monitored continuously and stopped at any time. What does that cost in samples? The question has a clean answer in the model that underlies the signs: i.i.d. $Z_t \sim \mathrm{Bern}(\frac{1+\Delta}{2})$ for an advantage $\Delta \in (0,1)$, tested against the exact null $\Delta = 0$. Write $\mathrm{KL}(\Delta) = \mathrm{KL}\big(\frac{1+\Delta}{2} \,\|\, \frac{1}{2}\big) = \frac{1+\Delta}{2}\log(1+\Delta) + \frac{1-\Delta}{2}\log(1-\Delta)$, which is $\Delta^2/2 + O(\Delta^4)$.

Proposition 8 (Expected stopping time) *Fix $\alpha \in (0,1)$ and $\Delta \in (0,1)$.*

(i) Lower bound. *Let τ be a stopping time with $\Pr_0[\tau < \infty] \le \alpha$ (a level-α sequential test that rejects at τ) and $\Pr_\Delta[\tau < \infty] = 1$, $\mathbb{E}_\Delta \tau < \infty$. Then $\mathbb{E}_\Delta[\tau] \ge \log(1/\alpha)/\mathrm{KL}(\Delta)$.*

(ii) Upper bound for the oracle bettor. *The wealth* $W_t = \prod_{u \le t}\big(1 + \Delta(2Z_u - 1)\big)$ *is a martingale under the null with growth rate* $\mathrm{KL}(\Delta)$ *under* Δ*, and* $\tau = \inf\{t : W_t \ge 1/\alpha\}$ *is a level-*α *test with*

$$\mathbb{E}_\Delta[\tau] \;\le\; \frac{\log(1/\alpha) + \log(1+\Delta)}{\mathrm{KL}(\Delta)}.$$

(iii) Grid mixture. *The uniform mixture of the constant-stake bettors with stakes* $\lambda_1, \dots, \lambda_K$ *has, for every* k *with* $g_\Delta(\lambda_k) > 0$, $\mathbb{E}_\Delta[\tau] \le \big(\log(K/\alpha) + \log(1 + \lambda_k)\big)/g_\Delta(\lambda_k)$*, where* $g_\Delta(\lambda) = \frac{1+\Delta}{2}\log(1+\lambda) + \frac{1-\Delta}{2}\log(1-\lambda)$.

Proof (i) Let $\Lambda_t = \sum_{u \le t} \log \frac{d\mathrm{Bern}((1+\Delta)/2)}{d\mathrm{Bern}(1/2)}(Z_u)$. Wald's identity [39] gives $\mathbb{E}_\Delta \Lambda_\tau = \mathrm{KL}(\Delta)\,\mathbb{E}_\Delta \tau$. Changing measure on $\{\tau = n\}$, where $d\Pr_0 / d\Pr_\Delta = e^{-\Lambda_n}$, $\alpha \ge \Pr_0[\tau < \infty] = \mathbb{E}_\Delta\big[e^{-\Lambda_\tau}\big] \ge e^{-\mathbb{E}_\Delta \Lambda_\tau}$ by Jensen, so $\mathbb{E}_\Delta \Lambda_\tau \ge \log(1/\alpha)$. (ii) W_t is a nonnegative martingale under the null with $W_0 = 1$, so Ville's inequality gives level α. Under Δ, $\mathbb{E}\log(1 + \Delta(2Z-1)) = \frac{1+\Delta}{2}\log(1+\Delta) + \frac{1-\Delta}{2}\log(1-\Delta) = \mathrm{KL}(\Delta) =: g$, and $\log W_t - gt$ is a martingale with bounded increments, so Wald's identity gives $\mathbb{E}\log W_\tau = g\,\mathbb{E}\tau$ once $\mathbb{E}\tau < \infty$, which holds since the drift is positive. Since $W_{\tau-1} < 1/\alpha$ and each factor is at most $1 + \Delta$, $\log W_\tau < \log(1/\alpha) + \log(1+\Delta)$. (iii) The mixture wealth is at least $W_t^{\lambda_k}/K$, so it crosses $1/\alpha$ no later than W^{λ_k} crosses K/α; apply (ii) with $\alpha \to \alpha/K$ and the stake λ_k, whose growth rate is $g_\Delta(\lambda_k)$. □

The oracle bettor knows Δ; a real bettor does not, and part (iii) and the simulations below price that ignorance. Part (i) shows that no sequential test, however clever, beats $\log(1/\alpha)/\mathrm{KL}$, and part (ii) that the oracle comes within $\log(1+\Delta)/\mathrm{KL}$ of it.

Against a fixed-sample test.

A two-sided sequential test runs two arms at level $\alpha/2$, so its oracle expected time is $2\log(2/\alpha)/\Delta^2\,(1 + o(1))$ as $\Delta \to 0$. The two-sided fixed-sample sign test at level α needs $n \approx (z_{1-\alpha/2} + z_{1-\beta})^2/\Delta^2$ pairs for power $1 - \beta$. At $\alpha = 0.05$ the constants are $2\log(40) = 7.4$ for the sequential oracle and 3.8, 7.9, 10.5 for power 0.5, 0.8, 0.9: the oracle's expected time equals the sample size of a fixed-sample test with about 78% power, and it can stop earlier when Δ is large and is valid under any stopping rule. Table 1 confirms the constants by simulation and adds the mixture over the ten stakes $0.05, 0.10, \dots, 0.50$ that the cohort analysis of Section 7 uses, which costs 1.1 to 1.4 times the oracle, more at small Δ.

This explains a result of Section 8.6. A paired fixed-sample sign test on the same pairs and the same score is about as powerful as VOUCH: the price of anytime validity is a constant factor of order one, not an order of magnitude. What the sequential test adds is that it may be monitored continuously and stopped at will without inflating the error, which the fixed-sample test may not. The cohort certificate of Section 7 is faster again, because its edge grows as the cohort is exhausted.

Table 1 Expected sample sizes in the sign model, $\alpha = 0.05$ (two arms at level 0.025). Wald: lower bound of Proposition 8(i); oracle: bound (ii) and simulated mean over 2,000 runs; grid: simulated mean of the 10-stake mixture over 500 runs; n_{50}, n_{80}, n_{90}: smallest n for which the exact two-sided binomial sign test at level 0.05 has the stated power.

Δ	KL	Wald	oracle $\leq$	oracle sim.	grid sim.	n_{50}	n_{80}	n_{90}
0.05	0.0013	2950	2989	2990	4225	1590	3234	4259
0.10	0.0050	737	756	749	952	409	809	1112
0.20	0.0201	183	192	186	211	103	204	284
0.30	0.0457	81	86	81	98	47	95	123
0.50	0.1308	28	31	30	34	17	33	43

7 When the cohort certificate arrives

VOUCH bounds the certification time of the cohort certificate through refutation (VOUCH Theorem 5): once enough zeros have been revealed, the null becomes infeasible and is rejected at any level. Refutation is exact but late, since it waits until the unrevealed remainder cannot carry the null. In practice the wealth process crosses $1/\alpha$ much earlier, by ordinary growth. This section bounds that time. The key observation is that under a uniform reveal order the betting edge of the without-replacement process is deterministic.

Lemma 9 (Deterministic edge) *Consider the upper certificate arm with $p_0 = \frac{1}{2} + \frac{\varepsilon}{2}$. Let the cohort carry j ones, $\hat{p} = j/m < p_0$, and $\delta = p_0 - \hat{p}$. Under a uniform reveal order, the boundary $p_0(t)$ of (1) and the conditional mean $r_t = \mathbb{E}[Z_{\pi(t)} \mid \mathcal{F}_{t-1}]$ satisfy, for every t and every history,*

$$g_t \;:=\; p_0(t) - r_t \;=\; \frac{m\delta}{m - t + 1}.$$

Proof With S_{t-1} ones revealed, $p_0(t) = (mp_0 - S_{t-1})/(m - t + 1)$ and, since the unrevealed remainder is a uniformly ordered set containing $j - S_{t-1}$ ones, $r_t = (j - S_{t-1})/(m - t + 1)$. Subtracting cancels S_{t-1}. □

The edge grows as the cohort is exhausted. On an i.i.d. stream with the same mean it would stay equal to δ; the growth is the finite-population advantage of the cohort target.

Theorem 10 (Certification time through wealth growth) *Fix the realised cohort, with j ones and $\delta = p_0 - \hat{p} > 0$ as in Lemma 9, so that the only randomness is the uniform reveal order. Run the upper arm with a uniform mixture of K constant stakes $\Lambda = \{\lambda_1, \ldots, \lambda_K\} \subset (0, \frac{1}{2}]$, refuting the null outright if $p_0(t) > 1$. Let H_n denote the harmonic numbers,*

$$A_t = \sum_{u \leq t} g_u = m\delta\,(H_m - H_{m-t}), \qquad G_t = \sum_{u \leq t} g_u^2 = m^2\delta^2 \sum_{k=m-t+1}^{m} k^{-2},$$

and for $\lambda \in \Lambda$

$$\Psi_t(\lambda) = \lambda A_t - \frac{\lambda^2(t/4 + G_t)}{2(1-\lambda)^2}, \qquad V_t(\lambda) = \frac{\lambda^2(t/4 + G_t)}{(1-\lambda)^2}, \qquad R_\lambda = \log\frac{1+\lambda}{1-\lambda}.$$

Then the stopping time τ *of the arm (wealth reaching* $1/\alpha$*, or refutation) satisfies, for every* $t < m$*,*

$$\Pr[\tau > t] \;\le\; \min_{\lambda\in\Lambda} \exp\Big(-\frac{x_t(\lambda)^2}{2\big(V_t(\lambda) + R_\lambda x_t(\lambda)/3\big)}\Big), \qquad x_t(\lambda) = \big(\Psi_t(\lambda) - \log(K/\alpha)\big)_+.$$

Proof Fix $\lambda \in \Lambda$ and let $X_u = \log\big(1+\lambda(p_0(u) - Z_{\pi(u)})\big)$, so that $\log W_t^\lambda = \sum_{u\le t} X_u$. Before refutation $p_0(u) \in [0,1]$, so $y_u = \lambda(p_0(u) - Z_{\pi(u)}) \in [-\lambda, \lambda]$. By Taylor's theorem with remainder, $\log(1+y) \ge y - y^2/(2(1-\lambda)^2)$ for $y \ge -\lambda$. Hence $\mathbb{E}[X_u \mid \mathcal{F}_{u-1}] \ge \lambda g_u - \lambda^2\,\mathbb{E}[(p_0(u) - Z_{\pi(u)})^2 \mid \mathcal{F}_{u-1}]/(2(1-\lambda)^2)$, and $\mathbb{E}[(p_0(u) - Z_{\pi(u)})^2 \mid \mathcal{F}_{u-1}] = r_u(1-r_u) + g_u^2 \le \frac{1}{4} + g_u^2$. Summing over $u \le t$, the compensator of $\log W_t^\lambda$ is at least $\Psi_t(\lambda)$, a deterministic quantity by Lemma 9. Further, $|\log(1+y)| \le |y|/(1-\lambda)$ for $|y| \le \lambda$, so the conditional variances of the X_u sum to at most $V_t(\lambda)$, and the centred increments are bounded below by $-R_\lambda$. After refutation, replace X_u by its compensator lower bound, so that the centred sequence remains a martingale difference sequence; this changes nothing on the event that refutation has not occurred by t. Freedman's inequality [40], applied to the negated increments, gives $\Pr[\log W_t^\lambda < \Psi_t(\lambda) - x] \le \exp\big(-x^2/(2(V_t(\lambda) + R_\lambda x/3))\big)$. Finally, the mixture wealth is at least W_t^λ/K, so on the event of no refutation by t we have $\{\tau > t\} \subseteq \{\log W_t^\lambda < \log(K/\alpha)\}$. Taking $x = x_t(\lambda)$ and minimising over Λ completes the proof. □

Reading the bound.

Since $1/(m-u+1) \ge 1/m$, $A_t \ge \delta t$, with equality only in the limit of an infinite cohort; the factor $\rho_t = m(H_m - H_{m-t})/t \approx 1 + t/(2m)$ is the finite-population speed-up. Ignoring G_t and the factor $(1-\lambda)^{-2}$ and optimising λ, the drift is about $2\delta^2 t\rho_t^2$, so the time scale is $t^* \approx \log(K/\alpha)/(2\delta^2\rho_{t^*}^2)$, against $\log(1/\alpha)/\mathrm{KL} \approx \log(1/\alpha)/(2\delta^2)$ for an i.i.d. stream. The original study found its cohort certificate faster than the super-population one at every tolerance and attributed this to the cap at cohort exhaustion, where the claim resolves deterministically. The bound shows that the speed-up starts well before exhaustion, through the edge that grows as the cohort runs out. The lower arm is the mirror statement with $\delta' = \hat{p} - (1 - p_0)$, and the two-sided certificate needs both arms. The theorem covers a constant-stake mixture. VOUCH's engine uses adaptive experts (online Newton step, Krichevsky–Trofimov); if the engine's wealth were a uniform average over its experts with the constant-stake grid among them, the same argument would apply with K replaced by the total number of experts, but we do not prove a bound for the adaptive experts themselves, and the numerical check below runs the constant-stake mixture, not the engine.

Scope.

The bound conditions on the realised cohort and is therefore not a sample-size rule fixed in advance: δ depends on the cohort's own advantage. It covers one arm of one score. A two-sided certificate over a class $\mathcal{F}$ is issued when all $2|\mathcal{F}|$ arms have crossed, so its time is the maximum of the arm times and, by a union bound, $\Pr[\tau_{\mathrm{cert}} > t]$ is at most the sum of the $2|\mathcal{F}|$ arm bounds.

Table 2 Certification time of the upper arm, $\hat{p} = \frac{1}{2}$, $\alpha = 0.05$, 2,000 uniformly random reveal orders per setting. "Bound $\leq \frac{1}{2}$" is the first checked time at which Theorem 10 bounds $\Pr[\tau > t]$ by one half; "fixed boundary" is the same bettor on an i.i.d. stream with the same mean.

m	ε	median τ	bound $\leq \frac{1}{2}$	ratio	fixed boundary	violations
512	0.2	119	346	2.9	142	0/32
1024	0.2	130	490	3.8	138	0/32
1024	0.1	370	810	2.2	527	0/32
2048	0.1	427	1226	2.9	532	0/32

Numerical check.

We compare the bound with the empirical tail of the same mixture bettor ($K = 10$ stakes, $0.05, 0.10, \ldots, 0.50$) over 2,000 uniformly random orders, at $\hat{p} = \frac{1}{2}$ and $(m, \varepsilon) \in \{(512, 0.2), (1024, 0.2), (1024, 0.1), (2048, 0.1)\}$ (Table 2). The bound is never violated at any of the 32 checked times per setting. It is conservative: the first time at which it falls below one half is 2.2 to 3.8 times the empirical median. The same bettor against a fixed boundary on an i.i.d. stream with the same mean is slower in every setting.

8 Experiments

The experiments ask five questions. How detectable are canaries, and can detectability be closed by design (Theorem 4)? Does an audit with semantic twins separate memorising from unlearned models on a real language model (Proposition 1)? When unlearning targets only the literal form, does the fact survive in paraphrase, and does auditing paraphrases change verdicts? What does an organic-matched cohort cost in power? How do membership-inference audits behave on the same runs? Every number below is produced by a script under `extension/experiments/`, stored under `extension/results/`, and turned into the tables and the figure of this paper by `make_tables.py` and `make_tables_ext.py` in `extension/manuscript/`. A test suite of eleven tests checks the generators and the exact null without model weights.

8.1 Setup

Data.

The organic records are fictitious biographies drawn from six sentence templates (birthplace and profession; study city; publication city; years in a city before a profession; a city guild prize; a workshop near a city) over fictitious first and last names, cities and professions; they are the keep and forget biographies of the VOUCH experiments. Each run uses 2,000 keep records, 200 organic forget records, a holdout of 500 organic records that are never trained, and a cohort of $m = 384$ twin pairs. Semantic twins cover five attributes (birthplace, profession, instrument, pet, colour) and are repeated 1, 2, 4 or 8 times by stratum; organic-matched twins (Section 5.4) appear once each. Belief scores average over paraphrased questions none of which is trained.

Table 3 Detecting canaries among organic records (400 canaries and 400 organic records, five-fold out-of-fold scores). A detector's advantage is a lower bound on the distinguishing advantage γ of Theorem 4. For reference, a detector with label-independent scores exceeds an advantage of 0.095 in 5% and 0.110 in 1% of 4,000 simulations at this sample size (`detect_null.py`).

	Perplexity features		Character n-grams		
Canary design	AUC	advantage	AUC	advantage	$\gamma \geq$
VOUCH random-string QA	0.998	0.973	1.000	1.000	1.000
semantic twin, literal	0.998	0.978	1.000	1.000	1.000
semantic twin, question–answer	0.892	0.605	1.000	1.000	1.000
organic-matched twin	0.521	0.052	0.442	0.102	0.102

Models and training.

The main model is Pythia-410M and the second size is Pythia-1B [3]; GPT-2 [41] is used for a pilot and for detector features. A LoRA adapter [42] of rank 16 on all linear layers is fine-tuned for 600 steps (batch 8, learning rate $2 \cdot 10^{-4}$, blocks of 96 tokens). Gradient ascent (GA) and NPO [5] act on a copy of the fine-tuned adapter with batch 4 and learning rate 10^{-4}; NPO uses a retain term on the keep data and the fine-tuned adapter as its reference. Retraining is a fresh adapter trained on the keep data only, so it satisfies the exact null; no unlearning is the positive control. The same step counts are used for both model sizes. All runs used single NVIDIA T4 GPUs.

Audit.

Verdicts use the VOUCH engine at $\alpha = 0.05$ and $\varepsilon = 0.2$. The runs draw their reveal order from local entropy, which is independent of the signs as Theorem 2 of VOUCH requires; the offline re-analyses below re-run the verifier on the stored score differences with 40 seeded reveal orders per cell, which averages the verdict over orders for the same model and is not a set of independent replications. REVOKED rejects the exact null, meaning that some declared score still depends on the coins, at any tolerance, while ISSUED certifies $|\Delta| < \varepsilon$ for every declared score.

8.2 Canary detectability

Theorem 4 bounds the gap between canary and organic advantage by twice the distinguishing advantage γ. We train two detectors to tell 400 canary in-twins from 400 organic records, both scored out of fold over five folds: a logistic regression on GPT-2 token-loss features (mean, maximum, median and spread of the token losses, and length), a trained version of the zero-knowledge perplexity filter of the original study, and a logistic regression on character 2–4-grams, which needs labelled canaries. The advantage $\max_\tau |\mathrm{TPR} - \mathrm{FPR}|$ of a detector is a lower bound on γ.

Table 3 compares VOUCH's random-string records, two forms of semantic twins built with their own templates, and organic-matched twins. The first three designs are detected with advantage one by the n-gram classifier, and the first two also by the perplexity features: random strings, a middle initial in every canary name and

Table 4 A selective provider guided by each detector: canary miss rate (the cohort advantage the audit sees) and fraction of organic records left un-unlearned, for three tolerances, with cross-fitted thresholds (400 canaries, 400 organic records). Random selection leaves a fraction equal to the miss rate.

Canary design	detector	$\varepsilon = 0.05$	$\varepsilon = 0.10$	$\varepsilon = 0.20$
VOUCH random-string QA	perplexity	0.05 / 0.99	0.10 / 0.99	0.21 / 1.00
VOUCH random-string QA	n-gram	0.06 / 1.00	0.11 / 1.00	0.20 / 1.00
semantic twin, literal	perplexity	0.05 / 1.00	0.10 / 1.00	0.21 / 1.00
semantic twin, literal	n-gram	0.05 / 1.00	0.10 / 1.00	0.21 / 1.00
semantic twin, question–answer	perplexity	0.05 / 0.57	0.10 / 0.68	0.20 / 0.79
semantic twin, question–answer	n-gram	0.05 / 1.00	0.11 / 1.00	0.20 / 1.00
organic-matched twin	perplexity	0.05 / 0.08	0.10 / 0.14	0.20 / 0.23
organic-matched twin	n-gram	0.05 / 0.03	0.10 / 0.04	0.20 / 0.12

attributes that never occur in organic data give the cohort away. For these designs Corollary 5 carries no information about organic records. Organic-matched twins bring both detectors close to the level of two organic samples. The perplexity detector's advantage, 0.052, is below the 95th percentile of the null reference. The n-gram detector's advantage, 0.102, is marginally above the 95th percentile (0.095) and below the 99th (0.110), and its AUC of 0.442 is below one half; refitting the same detector with permuted labels thirty times gives AUCs with mean 0.498 and minimum 0.440, and the same detector separating two independent organic samples gives 0.537, so the value is within the range of the detector's own noise (`detect_null.py`). Taken at face value, $\gamma \geq 0.10$ would still add 0.20 to the ladder of Corollary 5. These numbers are lower bounds on γ against two detector families. They do not show that no detector exists, and matching is easy here because we wrote the organic generator ourselves.

What a detector buys a selective provider.

Theorem 6 says that a provider that unlearns what a detector calls canary-like and leaves the rest un-unlearned passes the audit at tolerance ε when the detector misses a fraction ε of the canaries, and then leaves un-unlearned the fraction of organic records that the detector correctly calls organic-like. Table 4 measures this for our two detectors, with the threshold chosen on four folds and evaluated on the fifth. A provider that selects records at random leaves a fraction ε of organic records, which is the value that organic-matched canaries are meant to enforce.

Against VOUCH's random-string canaries and against the semantic twins built from their own templates, the provider passes the audit at $\varepsilon = 0.2$ while leaving all of the organic records un-unlearned, with either detector (at least 99% at every tolerance of the table); the certificate would say nothing about organic data. Against the question–answer form of semantic twins the perplexity detector leaves 79% at the same tolerance. Against organic-matched twins the leak is 0.23 for the perplexity detector and 0.12 for the n-gram detector at $\varepsilon = 0.2$, within the noise of the random value 0.2, as $\beta(\varepsilon) = \varepsilon$ predicts for indistinguishable canaries. The detectors are lower bounds on what a stronger detector could do.

Table 5 Semantic-twin audit on Pythia-410M with the full forget set ($m = 384$, three seeds). Realised advantages of the calibrated literal and paraphrase scores per seed, and the two-sided verdict over all scores (I issued, R revoked, U undetermined).

Subject	literal $\hat{\Delta}$	paraphrase $\hat{\Delta}$	$\varepsilon = 0.2$	$\varepsilon = 0.1$
no unlearning	+0.22, +0.31, +0.30	+0.20, +0.27, +0.26	R R R	R R R
GA (100 steps)	+0.04, +0.05, −0.09	+0.05, +0.05, −0.07	I I I	I I I
NPO (250 steps)	+0.04, −0.05, −0.05	+0.07, −0.07, −0.03	I I I	I I I
retrain	+0.02, +0.05, +0.01	+0.01, +0.11, +0.08	I I I	I U I

8.3 Auditing with semantic twins

Table 5 audits Pythia-410M when the forget set contains every form the in-twins were trained in.

Paraphrased questions that were never trained carry most of the literal advantage (0.24 against 0.28 on average for the calibrated scores), and the audit over all scores revokes the positive control and issues for retraining in every seed at $\varepsilon = 0.2$. On one retraining seed the calibrated paraphrase score alone would have revoked, which is within the 5% false-alarm rate per score that the exact null allows. A pilot with GPT-2 (128 pairs, 400 training steps, one seed) did not separate the controls: its positive control reached advantages of +0.25 (literal) and +0.17 (paraphrase) and was not revoked. Semantic twins need a model that generalises a stated fact to its paraphrases, which the smaller model barely does. With the full forget set, GA and NPO remove both advantages.

8.4 When unlearning targets only the literal form

The case a literal-only audit cannot see is unlearning that removes the trained surface form but not the fact. We study it in two designs, each sweeping the strength of unlearning over GA 5–25 steps and NPO 10–75 steps with eight seeds.

- *Incomplete forget set.* The in-twins are trained in the literal and one question–answer form, but the forget set contains only the literal form. This models a deletion request that names one record while the model also learned the fact from another record. Paraphrased questions resemble the question–answer form, which is never unlearned. Pythia-410M and Pythia-1B.
- *Literal only.* The in-twins are trained and unlearned in the literal form only, so the model can answer the paraphrased questions only by generalising from the statement. Whatever survives in paraphrase after unlearning has survived an unlearning objective that targeted the same record, which is the squeezing effect of Li et al. [2]. Pythia-410M.

The *retention* of a score is its advantage after unlearning divided by its advantage before unlearning. Because a single seed's pre-unlearning paraphrase advantage can be close to zero, which makes per-seed ratios unstable, we normalise each seed's advantages by the mean pre-unlearning advantage over seeds. We made this choice after

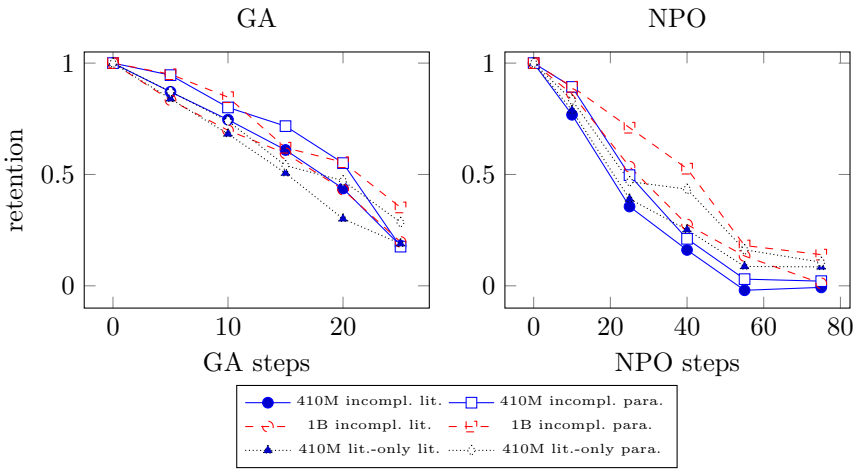


Fig. 1 Retention of the calibrated literal and the calibrated paraphrase advantage, normalised by the mean pre-unlearning advantage over eight seeds ($m = 384$). Filled markers: literal; open markers: paraphrase.

Table 6 Paraphrase minus literal retention, averaged over the whole grid of strengths within each seed; mean ± standard error over eight seeds, number of seeds with a positive value, and the Wilcoxon signed-rank p (smallest attainable value 0.008). The first row of each block is the like-for-like comparison.

Experiment	Scores compared	retention gap	seeds > 0	p
410M, incomplete	calibrated vs. calibrated	$+0.074 \pm 0.047$	6/8	0.195
410M, incomplete	raw vs. raw	$+0.077 \pm 0.033$	6/8	0.078
410M, incomplete	raw literal vs. calibrated paraphrase	$+0.172 \pm 0.075$	6/8	0.078
1B, incomplete	calibrated vs. calibrated	$+0.120 \pm 0.045$	6/8	0.078
1B, incomplete	raw vs. raw	$+0.083 \pm 0.041$	6/8	0.109
1B, incomplete	raw literal vs. calibrated paraphrase	$+0.163 \pm 0.051$	7/8	0.023
410M, literal only	calibrated vs. calibrated	$+0.080 \pm 0.059$	5/8	0.250
410M, literal only	raw vs. raw	$+0.065 \pm 0.082$	4/8	0.641
410M, literal only	raw literal vs. calibrated paraphrase	$+0.134 \pm 0.079$	6/8	0.195

observing the instability: our first analysis used per-seed ratios, and in the literal-only design it gives a gap of $+0.37 \pm 0.30$ (7 of 8 seeds positive, $p = 0.016$), a mean dominated by one seed whose pre-unlearning paraphrase advantage is close to zero; both versions are stored with the results. The test averages, within each seed, the difference between paraphrase and literal retention over the whole grid of strengths, and applies a Wilcoxon signed-rank test [43] to the eight per-seed values; seeds, not cells, are the unit of replication. The primary comparison is like for like: calibrated literal against calibrated paraphrase.

Table 7 Incomplete forget set: mean calibrated advantages over eight seeds and the fraction of runs (seeds × 40 reveal orders) revoked by the literal-only rule and by the union rule; A is the fraction of runs in which the literal rule issues a certificate and the union rule revokes. $\varepsilon = 0.2$, $\alpha = 0.05$.

	Pythia-410M					Pythia-1B				
Subject	lit.	para.	rev. lit.	rev. union	A	lit.	para.	rev. lit.	rev. union	A
none	+0.267	+0.219	1.00	1.00	0.00	+0.291	+0.209	1.00	1.00	0.00
GA 5	+0.232	+0.207	1.00	1.00	0.00	+0.243	+0.199	0.99	0.99	0.00
GA 10	+0.199	+0.175	0.92	0.97	0.05	+0.203	+0.177	0.87	0.82	0.01
GA 15	+0.163	+0.157	0.65	0.84	0.20	+0.173	+0.130	0.63	0.66	0.04
GA 20	+0.117	+0.120	0.40	0.58	0.19	+0.126	+0.116	0.48	0.45	0.02
GA 25	+0.048	+0.038	0.03	0.05	0.03	+0.057	+0.074	0.23	0.24	0.03
NPO 10	+0.205	+0.195	0.99	1.00	0.01	+0.251	+0.187	0.98	0.97	0.00
NPO 25	+0.095	+0.109	0.22	0.57	0.38	+0.156	+0.148	0.62	0.64	0.03
NPO 40	+0.043	+0.046	0.05	0.19	0.17	+0.080	+0.110	0.20	0.40	0.22
NPO 55	−0.005	+0.007	0.02	0.04	0.03	+0.039	+0.038	0.01	0.12	0.11
NPO 75	−0.002	+0.005	0.02	0.02	0.01	+0.003	+0.029	0.02	0.03	0.01

Paraphrase retention is slightly higher, but not significantly so.

Figure 1 shows paraphrase retention at or above literal retention at almost every strength in all three experiments, and Table 6 quantifies it. In the like-for-like comparison the gap is +0.07 to +0.12, positive in five or six of eight seeds, and not significant in any experiment ($p \geq 0.08$). Comparisons that mix raw and calibrated scores give larger gaps and, for Pythia-1B, a significant one, but they compare a raw score with a calibrated one, whose noise differs, and we do not rely on them; treating strength-by-seed cells rather than seeds as replicates would likewise overstate the evidence, since cells of one seed share its pre-unlearning model. In the literal-only design, where only generalisation can carry the fact to paraphrase, the gap is +0.08 with $p = 0.25$. We therefore do not claim to have demonstrated squeezing at this scale. The direction is the same in all three experiments, but they are not independent replications (they share seeds and canary manifests), and the size of the gap is within the seed-to-seed variation of eight runs.

Auditing paraphrases matters when the forget set is incomplete.

We compare three audit rules, re-running the verifier offline with 40 reveal orders per cell: the literal score alone at level α; a *union* of two separate audits, one on the literal score and one on the two calibrated paraphrase scores, each at level $\alpha/2$, which revokes if either revokes and issues only if both issue, so that its false-alarm probability under the exact null is at most α; and VOUCH's *mixture* engine over all six scores at level α. The union's literal arm uses the raw literal score and its paraphrase arm is itself a mixture over the two calibrated belief scores. With an incomplete forget set (Table 7), the union revokes far more often than the literal rule at intermediate strengths: after 25 NPO steps on Pythia-410M, 38% of runs end with a literal-only certificate that the union revokes, and after 40 NPO steps on Pythia-1B, 22%. In seed terms the literal rule issues while the union revokes in a majority of reveal orders in three of eight seeds at NPO 25 on Pythia-410M and in one of eight at NPO 40 on Pythia-1B; in the other seeds the fraction is at most 35%, so the averages are driven by a minority of seeds

Table 8 Literal-only design, Pythia-410M: mean calibrated advantages ± standard errors over eight seeds; fraction of runs revoked by the literal and the union rule and issued by the union rule (40 reveal orders per seed); and the membership-inference AUC of the organic forget biographies against the organic holdout (loss score).

Subject	literal $\hat{\Delta}$	paraphrase $\hat{\Delta}$	rev. lit.	rev. union	iss. union	organic AUC
none	$+0.292 \pm 0.017$	$+0.153 \pm 0.025$	1.00	1.00	0.00	0.55
retrain	$+0.011 \pm 0.010$	$+0.031 \pm 0.020$	0.02	0.01	0.99	0.51
GA 5	$+0.245 \pm 0.016$	$+0.133 \pm 0.023$	1.00	1.00	0.00	0.53
GA 10	$+0.199 \pm 0.018$	$+0.113 \pm 0.012$	0.96	0.96	0.04	0.51
GA 15	$+0.148 \pm 0.020$	$+0.083 \pm 0.013$	0.68	0.67	0.33	0.49
GA 20	$+0.088 \pm 0.026$	$+0.072 \pm 0.009$	0.25	0.25	0.75	0.50
GA 25	$+0.055 \pm 0.029$	$+0.044 \pm 0.019$	0.13	0.10	0.90	0.49
NPO 10	$+0.230 \pm 0.021$	$+0.128 \pm 0.018$	0.97	0.98	0.02	0.52
NPO 25	$+0.114 \pm 0.021$	$+0.072 \pm 0.011$	0.36	0.34	0.66	0.48
NPO 40	$+0.074 \pm 0.013$	$+0.066 \pm 0.008$	0.06	0.04	0.96	0.48
NPO 55	$+0.025 \pm 0.010$	$+0.025 \pm 0.018$	0.01	0.02	0.98	0.48
NPO 75	$+0.025 \pm 0.012$	$+0.016 \pm 0.014$	0.01	0.00	1.00	0.47

(per-seed values in `incomplete_stats.json`). The union is also slightly less powerful than the literal rule at some strengths on Pythia-1B, the price of splitting α. A literal-only certificate is not wrong in these runs: it states that the literal score's advantage is below ε, which it is. The union revokes because the paraphrase scores still depend on the coins, which a literal-only audit does not examine. In this design that is close to true by construction: the paraphrase scores resemble a form that was trained and never unlearned, and a larger declared class revokes more often. The experiment shows that a literal-only class is blind to such residue, not that the union is a better audit in general. Under no or light unlearning (none, GA 5, NPO 10) all rules revoke in at least 96.9% of runs, and after 75 NPO steps all revoke in at most 3%.

In the literal-only design the union adds nothing.

When the fact can reach paraphrase only through generalisation (Table 8), the paraphrase advantage before unlearning is about half the literal one and falls roughly in step with it. The literal and the union rule revoke at nearly identical rates at every strength, and the union issues on retraining in 99% of runs, so its false-alarm rate is within the nominal level. The paraphrase scores cost nothing in validity here and buy almost nothing in power. The practical lesson of this subsection is therefore narrower than squeezing: an audit that scores paraphrases catches knowledge the provider did not target, which an incomplete forget set leaves behind, and costs little when there is none.

8.5 Organic-matched twins in the audit

Table 9 runs the literal-only design with organic-matched twins, each appearing once in the corpus, on Pythia-410M with eight seeds.

The audit is valid and powerless. Retraining and every unlearning subject are revoked in at most 6% of runs, consistent with the nominal level. But so is the positive control: a record seen once in 600 training steps, whose twin differs from it in a single slot, leaves no detectable trace in the model, and the realised advantages are

Table 9 Organic-matched twins, literal-only design, Pythia-410M, eight seeds. Columns as in Table 8.

Subject	literal $\hat{\Delta}$	paraphrase $\hat{\Delta}$	rev. lit.	rev. union	iss. union	organic AUC
none	-0.036 ± 0.012	-0.033 ± 0.013	0.01	0.03	0.97	0.55
retrain	-0.035 ± 0.014	-0.036 ± 0.019	0.02	0.04	0.96	0.51
GA 10	-0.030 ± 0.012	-0.039 ± 0.014	0.02	0.03	0.97	0.52
GA 15	-0.036 ± 0.011	-0.027 ± 0.012	0.02	0.03	0.97	0.51
GA 20	-0.028 ± 0.016	-0.018 ± 0.014	0.03	0.03	0.97	0.50
GA 25	-0.020 ± 0.019	$+0.003 \pm 0.010$	0.03	0.02	0.98	0.50
NPO 25	-0.027 ± 0.016	-0.039 ± 0.019	0.03	0.03	0.97	0.50
NPO 40	-0.037 ± 0.017	-0.043 ± 0.022	0.03	0.04	0.96	0.47
NPO 55	-0.046 ± 0.017	-0.043 ± 0.019	0.05	0.05	0.95	0.46
NPO 75	-0.043 ± 0.018	-0.046 ± 0.018	0.06	0.06	0.94	0.45

within sampling noise of zero at every strength. (They are slightly negative on average, -0.035 ± 0.014 on retraining, 2.4 standard errors from zero. The rows of the table are not independent, since all subjects of a seed share its cohort and coins, and the exact null guarantees zero expectation for retraining by construction; the revocation rates stay within the nominal level.) The organic forget biographies, also seen once, are detected only weakly by a membership-inference AUC of 0.55. This is the cost that Corollary 5 hides: making the canaries indistinguishable from organic records removes the third term of the ladder, and at this training budget it also removes the memorisation signal the audit needs, at least at this training budget, the only one we tested. Repeating matched records would restore power but adds a signal of its own, since verbatim duplicates are rare in organic data. A verifier must choose a point on this trade-off, and the organic claim is only as good as the canaries are organic.

8.6 Comparison with membership-inference audits

The usual way to audit unlearning is a membership-inference test: score the forget set and a holdout set with an attack and ask whether the two are distinguishable, as in the PrivLeak metric of MUSE [9]. We run this test on the literal-only design (the runs of Table 8), with four membership scores computed on whole records: the loss, Min-K% Prob with $k = 20\%$ [20], the zlib ratio [19], and the loss minus the base model's loss ("ref"). A test declares leakage when the 95% bootstrap interval of the AUC excludes 0.5. Four pairings of members and non-members are compared: (a) canary in-twins against their ghost twins, the matched holdout that VOUCH's design provides; (b) canary in-twins against organic holdout records, the usual holdout of a benchmark; (c) the paraphrased questions of in-twins against those of ghost twins; and (d) the organic forget biographies against the organic holdout, the audit on the data users actually delete. To separate the contribution of pairing from that of sequential testing we add a paired fixed-sample test: a two-sided sign test at level 0.05 on the literal score differences of all 384 pairs, the same pairs and the same score that VOUCH uses, evaluated once at the end.

Table 10 shows four things.

An unmatched holdout makes the test meaningless. Pairing (b) reports leakage on every seed of every subject in the table, retraining included (and on seven of eight

Table 10 Fraction of seeds (VOUCH: of seeds × 40 reveal orders) in which each audit reports leakage, literal-only design, Pythia-410M, eight seeds. VOUCH columns are REVOKED rates of the literal and the union rule; "sign" is the fraction of seeds in which the paired fixed-sample sign test on the literal score rejects; membership-inference columns are fractions of seeds whose 95% bootstrap AUC interval excludes 0.5. Retraining is the exact null, so every positive entry in its row is a false alarm.

	VOUCH		sign	(a) canaries vs. ghosts				(b) vs. organic		(c) paraphrase		(d) organic	
Subject	lit.	union	lit.	loss	Min-K%	zlib	ref	loss	ref	loss	ref	loss	ref
none	1.00	1.00	1.00	1.00	1.00	1.00	0.75	1.00	1.00	0.00	0.00	0.38	0.12
retrain	0.02	0.01	0.00	0.00	0.00	0.00	0.00	1.00	1.00	0.00	0.00	0.12	0.00
GA 10	0.96	0.96	1.00	0.38	0.00	0.25	0.12	1.00	1.00	0.00	0.00	0.00	0.00
GA 15	0.68	0.67	0.88	0.00	0.00	0.00	0.00	1.00	1.00	0.00	0.00	0.25	0.00
GA 20	0.25	0.25	0.25	0.00	0.00	0.00	0.00	1.00	1.00	0.00	0.00	0.12	0.00
NPO 25	0.36	0.34	0.62	0.00	0.00	0.00	0.00	1.00	1.00	0.00	0.00	0.12	0.00
NPO 40	0.06	0.04	0.00	0.00	0.00	0.00	0.00	1.00	1.00	0.00	0.00	0.00	0.00
NPO 75	0.01	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00	0.00	0.00	0.25	0.12

seeds after 25 GA steps), with AUC 0.000 on retraining: the canaries and the organic holdout differ in distribution, so the test detects the difference between two kinds of record, not membership. Calibrating by the base model does not help. This is the confound of Theorem 4 seen from the side of the attack, and the reason Zhang et al. [23] require randomised inclusion; it matches the observations of Duan et al. [24] and Das et al. [25] for pre-trained models.

A matched holdout restores validity but not power. Pairing (a) uses the same records as VOUCH and is valid: no false alarm on retraining, and the positive control is detected in every seed by the loss, Min-K% and zlib scores (AUC 0.59). But under partial unlearning it falls silent long before VOUCH does: after 10 GA steps the loss test flags 38% of seeds where VOUCH revokes 96% of runs, and after 15 GA steps or 25 NPO steps it flags none where VOUCH revokes 68% and 36%. The paired fixed-sample sign test on the same pairs is at least as powerful as VOUCH at all strengths but one (for example 62% of seeds against 36% of runs after 25 NPO steps; after 40 NPO steps it rejects in no seed while VOUCH revokes 6% of runs), so the gap between VOUCH and the matched-holdout test comes from pairing and from scoring the target value rather than the whole record, not from sequential testing. An unpaired AUC compares in-twins with ghost twins across different entities, whose whole-record losses are dominated by the entities themselves; a paired test compares the two twins of one entity, which cancels that variation. Sequential testing costs some power against a test run once at the end, and buys the right to monitor continuously and stop at any time without inflating the error; the original paper measures what a fixed-sample test loses under optional stopping.

The paraphrase and organic audits are weak. Pairing (c) detects nothing, not even on the positive control, because a whole-record loss over a question and a one-word answer is dominated by the question. Pairing (d), the audit on the organic forget set itself, reaches an AUC of 0.55 on the positive control and flags it in three of eight seeds, with one false alarm in eight on retraining; at this training budget a single exposure leaves too little trace in an organic record for a fixed-sample test to see reliably.

9 Discussion and limitations

What the extension changes in practice.

Four recommendations follow for a verifier who audits with VOUCH. First, canaries that carry natural-language facts can be used without giving up the exact null, provided both twins are scored under the same prompt. Second, the declared class should contain calibrated paraphrase scores next to the literal score, audited as a union at level $\alpha/2$ each. Paraphrase scores never replaced the literal score in our experiments, since they are weaker on their own, but they reveal residue that an incomplete forget set leaves behind, which a literal-only class cannot see by construction, and they kept the error rate when the forget set was complete. Third, a verifier who wants the certificate to say something about organic records must pay for it twice: in detectability, by drawing canaries from the organic distribution, and in power, because such canaries are memorised no more than organic records are. Fourth, membership-inference tests against a holdout set should not be read as unlearning audits unless members and non-members come from one distribution by construction, as ghost twins do, and a paired test on twins is far more powerful than an unpaired one.

The distinguishing advantage cannot be certified.

Theorem 4 is conditional on γ, and data bound γ only from below. Our detectors are two cheap families; a provider with more compute could train stronger ones. What the theorem establishes is the direction of the burden: a provider that passes the audit on canaries while leaving organic data untouched has, in effect, built a canary detector, and the gap it obtains is bounded by that detector's advantage. The theorem covers recognition by content only. Recognition by provenance, for instance because canaries reach the provider through a separate channel, has to be prevented by the protocol, and closing the conditional altogether requires a computational assumption on the provider or a cryptographic attestation that the committed pipeline was run [28].

Detectability against power.

The organic-matched cohort in our experiments is both undetectable by our detectors and unmemorised by the model, and the second fact makes the first moot for an audit: a certificate issued on canaries that left no trace says that the pipeline leaves little trace of records seen once, which is true of the organic forget biographies as well, but it cannot distinguish a pipeline that unlearns from one that never learned. Larger training budgets, longer canaries spread over several slots, or repetition through paraphrased rather than verbatim copies are ways to buy back power, each with a detectability cost that would have to be measured. Generating matched canaries for real corpora, whose distribution the verifier sees only through samples, is the main open problem this work leaves.

Squeezing.

Our partial-unlearning experiments find the direction predicted by Li et al. [2], with paraphrase retention above literal retention in every experiment and at almost every strength, but an effect of 0.07 to 0.12 in retention that eight seeds do not separate from

zero. Comparisons that mix a raw literal score with a calibrated paraphrase score, or that treat cells rather than seeds as replicates, make the effect look larger and more significant than it is. Whether the effect grows with model size, training budget or a different unlearning objective, and whether it appears for facts the model also learned in pre-training, is open.

Scope of the experiments.

All facts are fictitious and all records are short biographies from one domain; fictitious facts guarantee that the base model knows neither value, at the price of realism. The models have at most one billion parameters and are fine-tuned with LoRA adapters for 600 steps; the same step counts are used for both sizes, so that Pythia-1B needs more unlearning steps for the same reduction is a statement about this configuration, not a tuned comparison. The membership-inference baselines are fixed-sample tests with bootstrap intervals, the form in which they are used in benchmarks; per-example calibrated attacks with shadow models [21, 22] were compared with VOUCH in the original paper and are not repeated here.

The theory.

Theorem 3 conditions on the training randomness and assumes bounded influence, which holds with $\kappa = \ell = 1$ under exact unlearning but is not verifiable under partial unlearning. The organic advantage Δ^o_ξ is a counterfactual quantity, the advantage the pipeline would show on organic data under randomised inclusion, not an observed rate. Theorem 10 is conservative by a factor of two to four, conditions on the realised cohort, and covers one arm of a constant-stake mixture rather than the adaptive experts of the VOUCH engine. Theorem 6 is sharp over leakage profiles, but whether a given training and unlearning pipeline realises the extremal profile is not addressed; it is a statement about what the audit data cannot rule out, not about what a particular provider does. Proposition 8 is stated for i.i.d. signs and an oracle stake. The signs of a cohort are drawn without replacement, which speeds certification up (Theorem 10), and the engine's adaptive experts pay a regret term that we do not bound.

10 Conclusion

VOUCH certifies forgetting on a committed cohort of planted canaries. This paper asked what such a certificate says about the data users delete, and what it can see. Semantic twins keep the exact null, so canaries can carry natural-language facts that have paraphrases. The certificate transfers to organic records at the cost of twice the provider's ability to tell canaries from organic records, a cost that existing canary designs make maximal and that organic-matched canaries remove, at the price of the memorisation signal the audit needs. Under partial unlearning the fact outlives its literal form only slightly at the scales we tested; an incomplete forget set is the larger practical risk, and an audit that scores paraphrases next to the literal form reveals it. Membership-inference tests against a holdout set either confound canaries with organic records or lack the power of a paired sequential test. The certificate remains a statement about a declared score class, a committed cohort and a provider

whose pipeline is attested or trusted, and the experiments show where each of these qualifications binds.

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Declarations

- **Funding.** No funding was received for this work.
- **Competing interests.** The author has no competing interests to declare.
- **Ethics approval and consent to participate.** Not applicable. The work uses no human participants and no personal data; all records in the experiments are fictitious and generated by the released code.
- **Consent for publication.** Not applicable.
- **Data availability.** All data are generated deterministically from committed seeds by the released code. Every result reported in this paper, including all per-seed records, is available in the repository listed below.
- **Materials availability.** Not applicable.
- **Code availability.** The code is available at https://github.com/vinhqdang/VOUCH-Verifiable-Online-Unlearning-Certification-via-Hypothesis-betting, in the directory `extension/`.
- **Author contribution.** Q.-V. D. conceived the work, developed the theory, implemented and ran the experiments, and wrote the manuscript.

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